

Recherche visuelle avec Microsoft Foundry et Cohere Embed 4

Recherche visuelle avec Microsoft Foundry et Cohere Embed 4.0



**global AI
community**
Virtuelle Francophone

Recherche visuelle avec Microsoft Foundry et Cohere Embed 4.0

Apprenez à construire des solutions de recherche visuelle en combinant Microsoft Foundry et le modèle Cohere Embed v4. Ce webinaire couvre l'intégration complète du pipeline : **génération d'embeddings vectoriels cohérents pour images et texte, indexation dans Azure AI Search, et requêtes multimodales depuis du texte ou des images.**

Une présentation de ce service à l'aide d'une démonstration sera réalisée lors de ce webinaire.



Serge RETKOWSKY

France AI Global Black Belt
@Microsoft



Paul PETON

AI Architect
@Expertime
AI MVP

LinkedIn Live

jeudi

19/02/2026

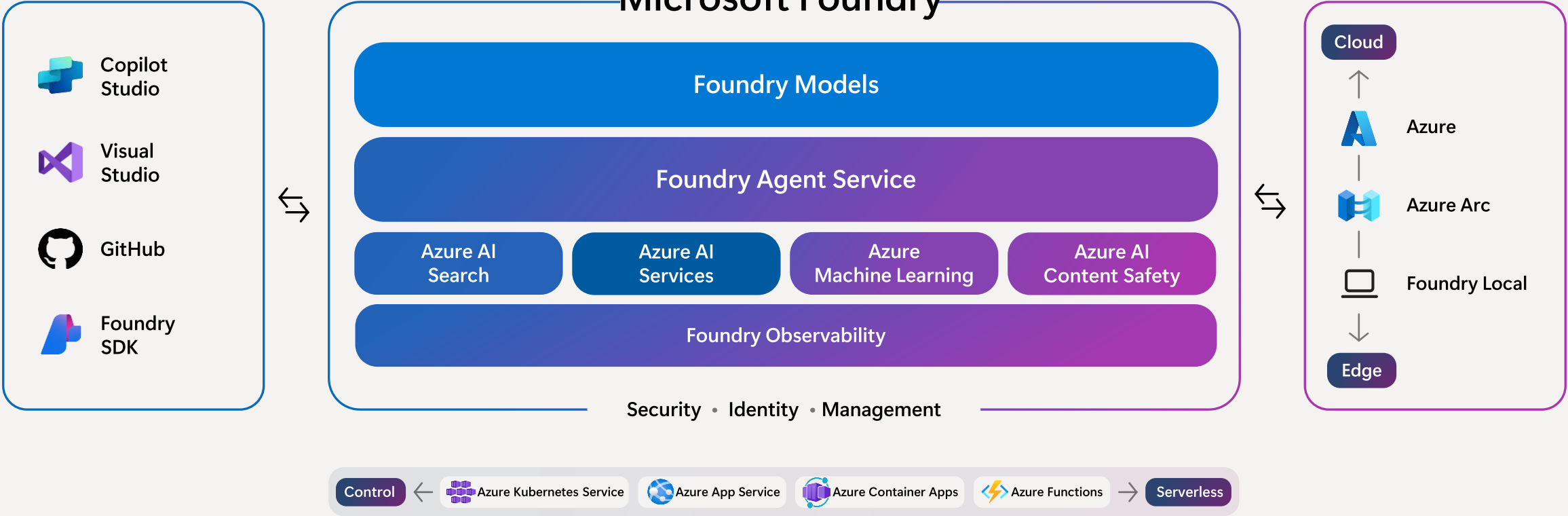
à partir de 12h

<https://www.linkedin.com/events/7428033909096996864>

Microsoft Foundry



Microsoft Foundry



Foundry models

Microsoft Foundry

/ demos-resource-project

Search with AI (Ctrl + K)

New Foundry

Home

Discover

Build

Operate

Docs

Overview

Models

Tools

Solution templates

Models (11307)

Search

Sort by Featured

Collections

Capabilities

Source

Inference tasks

Fine-tuning methods

Industry

claude-sonnet-4-6

Messages

gpt-5.2-chat

Chat completion, Responses

claude-opus-4-6

Messages

Kimi-K2.5

Chat completion

gpt-5.2-codex

Responses

DeepSeek-V3.2

Chat completion

DeepSeek-V3.2-Speciale

Chat completion

gpt-5.2

Chat completion, Responses

Cohere-rerank-v4.0-pro

Text classification

Cohere-rerank-v4.0-fast

Text classification

Kimi-K2-Thinking

Chat completion

gpt-5.1-codex-max

Responses

claude-opus-4-5

Messages

gpt-5.1

Chat completion, Responses

gpt-5.1-codex

Responses

DeepSeek-V3.1

Chat completion

gpt-5-chat

Chat completion, Responses

claude-haiku-4-5

Messages

model-router

Chat completion

Mistral-Large-3

Chat completion

sora-2

Video generation

embed-v-4-0

Embeddings, Summarization

claude-opus-4-1

Messages

grok-4

Chat completion

gpt-5.1-codex-mini

Responses

gpt-5-pro

Chat completion, Responses

gpt-5.1-chat

Chat completion, Responses

gpt-5

Chat completion, Responses

DeepSeek-V3-0324

Chat completion

gpt-4.1

Chat completion, Responses

Llama-4-Maverick-17B-128E-Instruct-F...

Chat completion

grok-4-fast-reasoning

Chat completion

gpt-4o-transcribe-diarize

Speech to text

FLUX.1-Kontext-pro

Text to image, Image to image

gpt-4.1-mini

Chat completion, Responses

FLUX-1.1-pro

Text to image

gpt-5-codex

Chat completion, Responses

o3

Chat completion, Responses

Microsoft Foundry / demos-resource-project

Search with AI (Ctrl + K)

New Foundry

Home

Discover

Build

Operate

Docs

Overview

Models

Tools

Solution templates

Models (13)

Search

Sort by Featured

Source: Cohere Clear filters

> Collections

> Capabilities

> Source (1)

☐ AI21 Labs 2

☐ Alibaba 1

☐ Anthropic 6

☐ Azure OpenAI 64

☐ Bayer 1

☐ Black Forest Labs 3

☐ BosonAI 1

☐ Bria 1

☒ Cohere 13

Cohere-rerank-v4.0-pro

Text classification

Cohere-rerank-v3.5

Text classification

Cohere-rerank-v3-multilingual

Text classification

Cohere-command-r

Chat completion

Cohere-rerank-v4.0-fast

Text classification

Cohere-embed-v3-multilingual

Embeddings

Cohere-rerank-v3-english

Text classification

embed-v-4-0

Embeddings, Summarization

Cohere-command-r-plus-08-2024

Chat completion

Cohere-embed-v3-english

Embeddings

cohere-command-a

Chat completion

Cohere-command-r-08-2024

Chat completion

Cohere-command-r-plus

Chat completion

Cohere Embed v 4

Microsoft Foundry Preview / demos-resource-project

Search with AI (Ctrl + K)

New Foundry Home Discover Build Operate Docs

Overview

Models

Tools

Solution templates

embed-v-4-0

Details Deployments Benchmarks License

Embed 4 transforms texts and images into numerical vectors

Cohere Direct from Azure Version: 6

Cohere's Embed 4 is a multilingual multimodal embedding model. It is capable of transforming different modalities such as images, texts, and interleaved images and texts into a single vector representation. Embed 4 offers state-of-the-art performance across all modalities (texts, images, interleaved texts and image) and in both English and multilingual settings.

Embed 4 supports a 128k context length and an images can have a maximum of 2MM pixels. Embed 4 is capable of vectorizing interleaved texts and images and capturing key visual features from screenshots of PDFs, slides, tables, figures, and more, thereby eliminating the need for complex document parsing. Embed 4 offers a variety of ways for compression both on the number of dimensions and the number-format precision. The model offers byte and binary quantization and matryoshka embeddings for further compression.

Deploy

Quick facts

Model provider	Cohere
Type	Embeddings, Summarization
Lifecycle	Generally available (GA)

Input type

image, text

Output type

image, text

Context

4096 output

Pricing

[View pricing](#)

Model ID

azureml://registries/azureml-cohere/models/embed-v-4-0/versions/6

Others who deployed this model also used

cohere-command-a

Chat completion

Cohere-embed-v3-multilingual

Embeddings

Cohere-embed-v3-english

Embeddings

How Vector Embeddings Work

Turning real-world data into numbers that capture meaning — the foundation of modern semantic search.

1

Multimodal Input

Embed 4 accepts text, images, and full documents (PDFs with tables, charts, handwriting) — all go through the same model.



Text



Image



Document

2

Dense Vector Output

Each input becomes a list of floating-point numbers — a point in high-dimensional space that encodes its semantic meaning.

1536 DIMENSIONS

0.0135, -0.0272, 0.0541, -0.0089, 0.0312,
-0.0447, 0.0198, -0.0156, ...



3

Semantic Similarity in Vector Space

Similar items cluster together. A photo of a **red dress** sits near a text description saying "elegant crimson evening gown" — even though they share no keywords. Dissimilar concepts (like electronics) land far away.

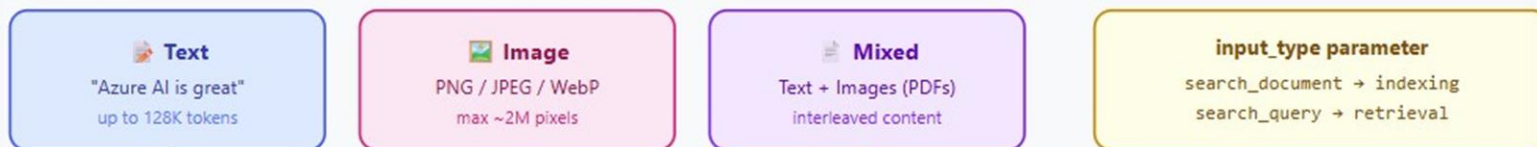
Cosine similarity measures the angle between two vectors. Values near **1.0** = very similar, near **0.0** = unrelated. This is how semantic search, recommendations, and anomaly detection all work under the hood.



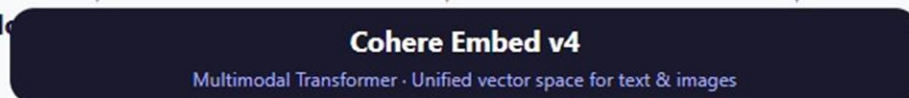
Cohere Embed v4 — How Embeddings Work

Multimodal embedding model · Text + Images → Vectors · 128K context · Matryoshka dimensions

① Inputs



② Model



③ Output: Embedding Vector



④ Matryoshka Dimensions

(choose precision vs. speed)



⑤ Embedding Types (Compression)



⑥ What You Can Do



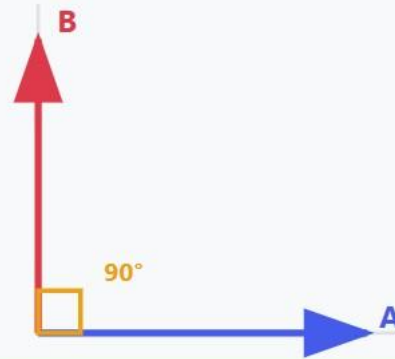
Cosine similarity

Cosine Similarity Explained

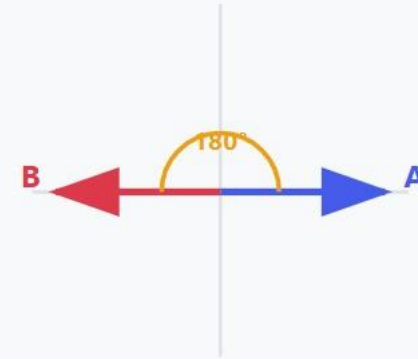
Measures the angle between two vectors — not their magnitude



$\cos(\theta) \approx 1$
Very Similar



$\cos(\theta) = 0$
No Relationship



$\cos(\theta) = -1$
Completely Opposite

$$\cos(\theta) = (A \cdot B) / (\|A\| \times \|B\|)$$

dot product divided by the product of magnitudes

Documentation

- Microsoft :

<https://techcommunity.microsoft.com/blog/azure-ai-foundry-blog/unlock-multi-modal-embed-4-and-multilingual-agentic-rag-with-command-a-on-azure/4404458>

- Model :

<https://ai.azure.com/catalog/models/embed-v-4-0>

<https://ai.azure.com/explore/models/embed-v-4-0/version/1/registry/azureml-cohere?tid=16b3c013-d300-468d-ac64-7eda0820b6d3>



Visual search article

- Medium article:
[Article](#)
- GitHub repo :
<https://github.com/retkowsky/visual-search-cohere-embed-v-4-0>

Building a Multimodal Visual Search Engine with Cohere Embed 4 and Microsoft Foundry



Serge Retkowsky · 7 min read · Dec 11, 2025



3



A complete guide to creating a multilingual fashion search application with text and image queries



Microsoft Foundry Cohere Embed 4 demos

1. Recherche visuelle

https://aka.ms/demo_cohere_embed_1

2. Détection des anomalies

https://aka.ms/demo_cohere_embed_2

3. Auto-tagging des images

https://aka.ms/demo_cohere_embed_3

Prochains évènements

**global AI
community**
Virtuelle Francophone

**Serge RETKOWSKY**
France AI Global Black Belt
@Microsoft

**Randa MEZRAG**
Consultante DATA & IA
@Finelog-Biseum

**Présentation du service
Azure Content Understanding**

Azure Content Understanding est un service Microsoft Foundry qui automatise l'extraction et la structuration de contenu multimodal (documents, images, vidéos, fichiers audios) sans labélisation complexe.

Une présentation de ce service à l'aide de démonstrations sera réalisée lors de ce webinar.

 **Live**
jeudi
12/03/2026
à partir de 13h

**global AI
community**
Virtuelle Francophone

**Serge RETKOWSKY**
France AI Global Black Belt
@Microsoft

**Alibek JAKUPOV**
France AI Global Black Belt
@Microsoft

**Nicolas ROBERT**
Profession
@Cellenza
AI MVP

**Présentation des accélérateurs
Citadel et Spec2Cloud**

Citadel et Spec2Cloud sont deux accélérateurs Microsoft IA. Citadel fournit une **gouvernance centralisée, la conformité et l'observabilité** pour déployer des agents IA sécurisés et contrôlés à l'échelle. Spec2Cloud accélère le **développement d'applications et d'agents IA** via des modèles de spécifications.

Ensemble, ils résolvent le dilemme vitesse vs contrôle : Spec2Cloud vous fait développer vite, Citadel vous assure de le faire en sécurité et conformité. Ces solutions offrent une base pour **transformer rapidement les idées en agents IA gouvernés et production-ready**.

 **Live**
jeudi
19/03/2026
à partir de 13h

<https://lnkd.in/ewUig486>

https://lnkd.in/eUe_bBrz

<https://www.linkedin.com/company/global-ai-community-france/posts/?feedView=all>

Replays Global AI Community



[\(7\) Global AI Community Virtuelle Francophone: Posts | LinkedIn](#)

Merci